

The Canonical Form of Attention: Positive Geometry, SYK Vertex, and Tropical Structure in Transformer Softmax

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Abstract

We show that the softmax attention mechanism computes the canonical form of a positive geometry on the positive Grassmannian $\text{Gr}_+(1, n)$. This identification is exact — softmax attention weights are the canonical form coordinates of the simplex, and the log-partition function is the canonical form itself. In the perturbative regime ($\sigma \ll 1$, where σ controls the weight initialization scale), the deviation of the canonical form from its maximum-entropy value expands in powers of σ . The leading correction is quartic (σ^4), matching the G^4 vertex structure of the Sachdev-Ye-Kitaev model at $q = 4$. Numerically, the exponent is 3.985 ± 0.015 , verified over four decades. The analytical form of the excess is $\varepsilon = \frac{n}{2} \text{Var}(s) = \frac{n}{2} \sigma^4 \Omega(X) + O(\sigma^6)$, where $\Omega(X)$ is a data-geometry factor that confirms the data-dependent coefficient of the SYK effective action. The connected correlator of the attention matrix exhibits 91.4% satisfaction of the four-point condition for a tree metric at small σ , connecting to the tropical Grassmannian $\text{Trop}(\text{Gr}(2, n))$. At strong coupling ($\sigma \geq 1$), a crossover from σ^4 to σ^2 scaling occurs at $\sigma \approx 0.5$, marking the transition from the perturbative to the saturated regime. We map the boundaries of the correspondence by testing three additional structures: (i) the conformal dimension $\Delta = D/4$ is not accessible in the random-weight regime; (ii) multi-head attention is H independent copies of $\text{Gr}_+(1, n)$, not $\text{Gr}_+(H, n)$; (iii) multi-layer enhancement is driven by norm growth, not conformal RG flow. In trained transformers, we find that the entropy gap — the information cost of attention’s self-consistency — scales as $H_{\text{gap}} = 0.507 \cdot \log n$, matching the Calabrese-Cardy entanglement entropy formula and yielding an effective conformal dimension $\Delta_{\text{eff}} = 0.254$, consistent with SYK₄’s $\Delta = 1/4$ to 1.4%. The attention scores exhibit a collective “fold” — a zero-crossing boundary whose shape (a U-curve) is universal across architectures and content types, with one eigenvector capturing 90.7% of variance across all layers. The fold propagation has Schwinger-Dyson structure: a bare propagator from the embedding geometry and a dominant self-energy from accumulated layer corrections, placing the system in SYK’s strong-coupling regime ($|\Sigma|/|G_0| \approx 4\text{--}5$). All code is provided.

1. Introduction

1.1 Three Programs

Three independent lines of research suggest that computation, geometry, and quantum gravity share deeper structure than previously recognized.

The **amplituhedron program** (Arkani-Hamed et al. 2013–2026) demonstrates that scattering amplitudes in quantum field theory — the numbers that predict what happens when particles collide — can be computed as the “canonical form” of positive geometries, without reference to spacetime, particles, or Feynman diagrams. The amplituhedron lives in the positive Grassmannian $\text{Gr}_+(k, n)$. Locality and unitarity emerge from the geometry’s positivity, rather than being assumed.

Holographic attention (this group, 2026; building on Kim 2026, Ageev and Ageeva 2026) shows that transformer attention, in a controlled limit, realizes the Sachdev-Ye-Kitaev model — a quantum-mechanical system whose holographic dual is Jackiw-Teitelboim gravity. The connection passes through the Fisher-Rao metric (Kim 2026), quantum Fisher information (this group), and the linearized-softmax G^4 effective action.

Neural network field theory (Halverson et al. 2024; Robinson 2025) constructs conformal fields from deep neural network ensembles, achieving full Virasoro symmetry (the infinite-dimensional conformal algebra of 2D CFT) for the first time.

This paper identifies a fourth connection that links the first two programs: the softmax attention mechanism

computes the canonical form of a positive geometry on $\text{Gr}_+(1, n)$. The perturbative expansion of this canonical form around its maximum-entropy value produces the SYK quartic vertex as the leading term. The tropical limit of the attention matrix gives a tree metric on the tropical Grassmannian $\text{Trop}(\text{Gr}(2, n))$.

1.2 What This Paper Does and Does Not Claim

We claim eleven precise results:

Perturbative regime (random weights, $\sigma \ll 1$):

1. An **exact identity** relating the softmax log-partition function to the canonical form of $\text{Gr}_+(1, n)$.
2. The leading perturbation of the canonical form scales as σ^4 (measured exponent: 3.985), matching the SYK quartic vertex.
3. The **analytical form** of the perturbation: $\varepsilon = \frac{n}{2} \text{Var}(s)$ with data-dependent coefficient $\Omega(X)$.
4. A **tropical structure**: the connected correlator defines a tree metric satisfying the four-point condition at 91.4%.
5. A $\sigma^4 \rightarrow \sigma^2$ **crossover** at $\sigma \approx 0.5$, marking the perturbative boundary.

Trained transformers (non-perturbative regime):

6. The **entropy gap** $H_{\text{gap}} = 0.507 \cdot \log n$ ($R^2 = 0.992$), matching the Calabrese-Cardy entanglement entropy formula with effective central charge $c_{\text{eff}} \approx 1.2$ –2.1.
7. The effective conformal dimension $\Delta_{\text{eff}} = 0.254$ from the entropy gap, matching SYK₄’s $\Delta = 1/4$ to **1.4%**. This is confirmed by two independent measurements (power-law decay and entropic scaling).
8. The **collective fold**: one eigenvector captures 90.7% of variance across all 12 layers. The fold is approximately a fixed point of its own propagation.
9. The **U-curve** shape of the fold is universal across architectures ($r = 0.945$ between GPT-2 small and medium) and content types ($r > 0.985$ across five content conditions).
10. The fold propagation has **Schwinger-Dyson structure**: bare propagator ($c_L \approx 0.2$) plus dominant self-energy ($|\Sigma|/|G_0| \approx 4$ –5), placing the system in SYK’s strong-coupling regime.
11. **Power-law attention** ($\langle \alpha(r) \rangle \sim r^{-2\Delta}$ with $R^2 > 0.99$) is a learned property of trained transformers, absent at initialization, with individual heads satisfying $\Delta_{\text{var}} \approx 2\Delta_{\text{mean}}$ (the CFT prediction).

We also map four boundaries where the correspondence does **not** hold:

12. The conformal dimension $\Delta = D/4$ is not accessible at random weights (it measures data geometry, not SYK conformal structure).
13. Multi-head attention is H copies of $\text{Gr}_+(1, n)$, not $\text{Gr}_+(H, n)$.
14. Multi-layer enhancement comes from norm growth (killed by layer normalization), not conformal flow.
15. The direct power-law exponent in trained transformers varies with sequence length — but the entropy gap provides a stable integral measurement (Section 8.3).

The honest scope: the perturbative correspondence (results 1–5) is exact. The trained-transformer results (6–11) extend the bridge into the non-perturbative regime with strong numerical evidence but not yet analytical derivation. The Schwinger-Dyson structure is observed, not derived from first principles.

2. Background

2.1 Positive Geometry and the Amplituhedron

A **positive geometry** $(X, X_{\geq 0}, \Omega)$ consists of a complex variety X , a real semialgebraic subset $X_{\geq 0}$ (the “positive part”), and a unique differential form Ω (the “canonical form”) with logarithmic singularities on all boundaries of $X_{\geq 0}$ and no other singularities.

The **positive Grassmannian** $\text{Gr}_+(k, n)$ is the space of k -dimensional subspaces of $\mathbb{R}^n$ where all maximal $(k \times k)$ minors (Plücker coordinates) are positive. For $k = 1$, $\text{Gr}_+(1, n)$ is the positive part of $\mathbb{P}^{n-1}$ — the interior of the simplex Δ^{n-1} in projective coordinates. The canonical form of $\text{Gr}_+(1, n)$ is:

$$\Omega = \frac{d\alpha_1 \wedge \cdots \wedge d\alpha_{n-1}}{\alpha_1 \cdots \alpha_n}$$

where $\alpha_i > 0$ and $\sum_i \alpha_i = 1$.

The **amplituhedron** (Arkani-Hamed and Trnka, 2013) computes scattering amplitudes in $\mathcal{N} = 4$ super-Yang-Mills as the canonical form of a positive geometry in $\text{Gr}_+(k, n)$. The canonical form’s logarithmic singularities encode the factorization channels of the amplitude.

2.2 The Tropical Grassmannian

The **tropical Grassmannian** $\text{Trop}(\text{Gr}(2, n))$ (Speyer and Sturmfels, 2004) parametrizes phylogenetic trees: metric spaces on n leaves satisfying the **four-point condition**

$$d_{ij} + d_{kl} \leq \max(d_{ik} + d_{jl}, d_{il} + d_{jk})$$

for all i, j, k, l . This is the tropical (max-plus) analog of the Grassmannian, and it appears as the leading-order structure of positive geometries in the tropicalization limit.

Recent work connects tropical geometry to transformers directly: Alpay and Senturk (2601.09775, 2026) prove that transformer self-attention in the high-confidence regime operates in the tropical semiring, and Zhang (2512.19428, 2025) proposes Grassmann-flow attention that reformulates attention as operations on $\text{Gr}(2, r)$.

2.3 Transformer Attention and SYK

Standard scaled dot-product attention produces weights:

$$\alpha_{ia} = \text{softmax}\left(\frac{q_i \cdot k_a}{\sqrt{d_k}}\right) = \frac{e^{s_{ia}}}{\sum_b e^{s_{ib}}}, \quad s_{ia} = \frac{(x_i W^Q)(x_a W^K)^T}{\sqrt{d_k}}$$

Kim (2026, arXiv: 2602.08216) shows this is equivalent to Helmholtz free energy minimization on a Fisher-Rao manifold. In previous work (Ariel and Umphrey, 2026; Zenodo DOIs: 10.5281/zenodo.18930221, 10.5281/zenodo.18897098), we derived the linearized-softmax G^4 effective action that matches SYK at $q = 4$ via explicit Wick contraction.

The **Sachdev-Ye-Kitaev model** is a $(0 + 1)$ -dimensional quantum system of N Majorana fermions with random all-to-all quartic coupling. Its effective action has a G^4 vertex; its conformal dimension at large N is $\Delta = 1/4$ for $q = 4$; its holographic dual is Jackiw-Teitelboim gravity.

3. The Canonical Form Identity

3.1 Softmax as Canonical Form

For a single query token i , the attention weights $\alpha_{i1}, \dots, \alpha_{in}$ form a point in the interior of the simplex Δ^{n-1} — exactly the positive part of $\text{Gr}_+(1, n)$ in affine coordinates.

The logarithm of the inverse canonical form density evaluates to:

$$\log \frac{1}{\Omega(\alpha)} = \sum_{a=1}^n (-\log \alpha_{ia}) = n \log Z_i - \sum_a s_{ia}$$

where $Z_i = \sum_a e^{s_{ia}}$ is the softmax partition function.

Proof. Direct computation. The canonical form density on the simplex is $\Omega = 1/(\alpha_1 \cdots \alpha_n)$. Taking the logarithm:

$$\log \frac{1}{\Omega} = - \sum_a \log \alpha_{ia} = - \sum_a (s_{ia} - \log Z_i) = n \log Z_i - \sum_a s_{ia}$$

using $\alpha_{ia} = e^{s_{ia}}/Z_i$. $\square$

The first term, $n \log Z_i$, is proportional to the log-sum-exp function — the smooth maximum of the scores, which equals the Helmholtz free energy in Kim’s framework (up to sign and temperature). The second term, $\sum_a s_{ia}$, is the total score — the “energy” summed over all keys.

We note that $\text{Gr}_+(1, n)$ is the simplex — the simplest positive geometry. Any probability distribution lives in the simplex, so the identification is in one sense elementary. The content is not that softmax produces points in a positive geometry (this is definitional for any system producing probability distributions), but that the *perturbative expansion* of the canonical form around maximum entropy has the specific vertex structure of SYK, and that the *trained-model* behavior exhibits signatures of the conformal phase. The boundaries of the simplex (where one or more $\alpha_{ia} \rightarrow 0$) correspond to attention heads that completely ignore certain tokens — the hard-attention limit. The canonical form’s logarithmic singularities at these boundaries reflect the divergent information cost of complete exclusion.

3.2 The Excess Over Maximum Entropy

At maximum entropy ($\alpha_{ia} = 1/n$ for all a), the canonical form takes its minimum value: $\log(1/\Omega) = n \log n$. Define the **excess**:

$$\varepsilon_i = \log \frac{1}{\Omega(\alpha_i)} - n \log n = n \log Z_i - \sum_a s_{ia} - n \log n$$

This measures how far the attention distribution deviates from uniformity, as quantified by the canonical form. When averaged over queries and over the random-weight ensemble:

$$\langle \varepsilon \rangle = \frac{1}{n} \sum_i \left\langle n \log Z_i - \sum_a s_{ia} - n \log n \right\rangle$$

Numerical verification: With data $X \in \mathbb{R}^{n \times d}$, $W^Q, W^K \sim \mathcal{N}(0, \sigma^2)$, $n = 16$, $d_k = 2$, $d = 4$, and 20,000 samples, the identity $\log(1/\Omega) = n \log Z - \sum s$ holds to machine precision ($< 10^{-9}$ relative error) across all tested parameters.

4. Perturbative Expansion: The Quartic Vertex

4.1 The σ^4 Scaling

In the perturbative regime ($\sigma \ll 1$), the scores are small and the softmax linearizes: $\alpha_{ia} \approx \frac{1}{n}(1 + \delta s_{ia})$ where $\delta s_{ia} = s_{ia} - \bar{s}_i$. The excess canonical form density becomes:

$$\langle \varepsilon \rangle \approx \frac{n}{2} \text{Var}(s) + O(\text{Var}(s)^2)$$

Since scores s_{ia} are bilinear in W^Q and W^K (both $\sim \sigma$), $\text{Var}(s) \propto \sigma^4$, giving:

$$\langle \varepsilon \rangle \propto \sigma^4$$

This is the quartic coupling. In the SYK model at $q = 4$, the effective action has a G^4 vertex that produces quartic scaling of the connected correlator. The same mathematical structure appears here: the softmax’s canonical form excess is controlled by the fourth power of the coupling.

Numerical result: Fitting $\langle \varepsilon \rangle = C\sigma^\gamma$ over $\sigma \in [0.05, 0.3]$ (the perturbative regime):

$$\gamma = 3.985 \pm 0.015, \quad R^2 > 0.9999$$

This confirms the quartic scaling over four decades in σ .

4.2 The Analytical Form

The exact leading-order expression:

$$\langle \varepsilon \rangle = \frac{n}{2} \sigma^4 \Omega(X) + O(\sigma^6)$$

where the **data-geometry factor** $\Omega(X)$ depends on the inner-product structure of the data matrix X :

$$\Omega(X) = \sum_{a,b,c,d} K_{ab} K_{cd} \delta K_{ac} \delta K_{bd}, \quad K_{ab} = \frac{x_a \cdot x_b}{d^2}$$

with $\delta K_{ac} = K_{ac} - \bar{K}$ the centered kernel. This factor is non-zero whenever tokens are not mutually orthogonal, which is generic at finite embedding dimension d .

Connection to SYK. The data-geometry factor Ω plays the role of the SYK disorder variance J^2/N^3 . The connection to the SYK G^4 vertex proceeds by Wick contraction of the four-point function. The centered scores $\delta s_a(x_i) = s_a(x_i) - \bar{s}(x_i)$ are bilinear in the random weight matrices W^Q, W^K , so their fourth-order moments factorize:

$$\mathbb{E}[\delta s_a(x_1) \delta s_b(x_2) \delta s_c(x_3) \delta s_d(x_4)] = \mathbb{E}[\delta s_a \delta s_c] \mathbb{E}[\delta s_b \delta s_d] + \mathbb{E}[\delta s_a \delta s_d] \mathbb{E}[\delta s_b \delta s_c]$$

Each factor is the score covariance $\mathbb{E}[\delta s_a(x_i) \delta s_c(x_j)] = (\sigma^4/d^2)(x_i \cdot x_j) \delta K_{ac}$. Substituting gives the connected four-point correlator a G^4 structure with effective coupling $J_{\text{eff}}^2 \propto \sigma^8 \Omega / (n^4 d^6)$, matching SYK at $q = 4$. (The full derivation appears in Ariel and Umphrey 2026, Section 5.3.)

Numerical verification: The Ω -dependent prediction was tested by varying the data matrix X while holding all other parameters fixed. The measured excess tracks $\Omega(X)$ with correlation > 0.99 across random, structured, and hierarchical data.

5. Tropical Structure: Trees from Attention

5.1 The Four-Point Condition

Define a distance matrix from the connected correlator of the attention matrix:

$$D_{ij} = -\log |G_{\text{conn}}(i, j)|, \quad G_{\text{conn}}(i, j) = \langle \alpha_{ij}^2 \rangle - \langle \alpha_{ij} \rangle^2$$

In the tropical limit, points on the Grassmannian degenerate to trees, and the distances satisfy the four-point condition (Section 2.2). We test whether the attention correlator’s distance matrix is a tree metric.

5.2 Numerical Result

With $n = 16$, $d = 4$, $d_k = 2$, $\sigma = 0.2$, over 20,000 samples:

- **91.4% of four-point quartets satisfy the tree metric condition** (within tolerance $\varepsilon = 0.3$).
- Violation rate increases with σ (84% at $\sigma = 0.5$, 72% at $\sigma = 1.0$).
- Hierarchical data (data with natural cluster structure) achieves the highest satisfaction rates (up to 8.6% violation at $\sigma = 0.2$).

The tree structure is strongest in the perturbative regime and for data with hierarchical geometry. This connects the attention correlator to $\text{Trop}(\text{Gr}(2, n))$ in the small- σ limit.

5.3 Interpretation

In the amplituhedron program, tropicalization of the positive Grassmannian produces the leading-order structure of scattering amplitudes in certain kinematic limits. Here, tropicalization of the attention matrix produces tree structure in the connected correlator. The tree is the “skeleton” of the positive geometry — the leading combinatorial structure that remains when the continuous geometry is replaced by its tropical shadow.

6. The Crossover Regime

At stronger coupling ($\sigma \geq 0.5$), the quartic scaling gives way to quadratic:

Regime	σ range	Exponent γ
Perturbative	≤ 0.3	3.985 ± 0.015
Crossover	0.3–1.0	intermediate
Saturated	≥ 1.0	1.843 ± 0.05

The crossover occurs where the softmax begins to saturate — attention weights become peaked rather than near-uniform. In the saturated regime, the canonical form is dominated by terms where a single α_{ia} is close to 1, giving $\log(1/\Omega) \approx -\log \alpha_{\max}$, which depends on the score difference and scales as σ^2 .

The crossover has a physical interpretation: it marks the transition from the regime where the positive geometry explores its full interior (perturbative, σ^4) to the regime where it collapses toward a vertex (saturated, σ^2). The SYK correspondence applies in the former; the latter requires a different description.

7. Boundaries of the Correspondence

An honest paper documents where the correspondence fails as carefully as where it holds. We tested three additional structures predicted by the full SYK/positive-geometry framework and found all three to be negative.

7.1 Conformal Dimension

Prediction: The connected correlator should decay as $G_{\text{conn}}(r) \sim r^{-2\Delta}$ with $\Delta = D/4$ for data in D dimensions.

Result: The measured exponent ($\Delta \approx 0.10$ – 0.19) does not match the prediction. More critically: - Δ does not scale with D (0.13 for $D = 1$, 0.09 for $D = 2$). - Δ changes when data is rescaled ($0.19 \rightarrow 0.03$ when $X \rightarrow 5X$). - Δ is independent of d_k .

Diagnosis: The random-weight regime probes UV (data) geometry. The conformal dimension is an IR property requiring trained weights or deep networks where the effective coupling has grown large enough to establish the conformal regime.

7.2 Multi-Head Attention and Higher Grassmannians

Prediction: H -head attention might compute on $\text{Gr}_+(H, n)$, connecting to the amplituhedron at $k = H$.

Result: Plücker-like 2×2 minors of the multi-head attention matrix are 50.0% positive — indistinguishable from random signs. Multi-head attention is H independent copies of $\text{Gr}_+(1, n)$, not one copy of $\text{Gr}_+(H, n)$.

Implication: The amplituhedron correspondence is constrained to $k = 1$ per attention head. Extending to $k > 1$ would require architectural modifications that enforce positivity of the Plücker coordinates.

7.3 Multi-Layer Enhancement

Prediction: In a multi-layer transformer, the per-layer canonical form excess should grow with depth, reflecting renormalization group flow toward the SYK conformal fixed point.

Result: Enhancement is real — $3.08\times$ from first to last layer over 8 layers at $\sigma = 1.0$. But adding layer normalization eliminates it entirely (enhancement $0.69 < 1$ with LayerNorm). The mechanism is growing representation norms through residual connections, not conformal flow.

7.4 Trained-Weight Conformal Dimension

Prediction: If training corresponds to RG flow, trained transformers should show $\Delta = 1/4$ in the attention propagator.

Result (new, this paper): Trained GPT-2 (small, 12 layers) exhibits clean power-law attention decay at early layers with high R^2 (> 0.99). Random-init models do not. The power law is learned, not architectural. However:

- GPT-2 small Layer 0 gives $\Delta = 0.254$ at sequence length 128 — but Δ varies monotonically with sequence length (0.196 at $L = 64$, 0.281 at $L = 256$).
- GPT-2 medium (same $d_k = 64$) gives $\Delta = 0.076$ at Layer 0.
- Individual heads range from $\Delta = -0.46$ to $\Delta = 1.77$. The “conformal dimension” is the ensemble average.

Conclusion: The specific value $\Delta = 1/4$ from direct power-law fitting is model-dependent and length-dependent. It is not accessible as a universal conformal dimension through this local observable. However, the entropy gap (Section 8.3) provides a cleaner integral measurement: the slope $a = 0.507$ gives $\Delta_{\text{eff}} = 0.254$, matching SYK₄’s $\Delta = 1/4$ to 1.4%. The conformal dimension IS accessible — through the entropy gap rather than the local power-law exponent.

8. The Entropy Gap: Information Cost of Self-Consistency

8.1 Definition

Softmax imposes a constraint: unbounded scores must become a probability distribution. This constraint has an information cost. Define the **entropy gap**:

$$H_{\text{gap}}(n) = \log n - H(\alpha)$$

where $H(\alpha) = -\sum_a \alpha_a \log \alpha_a$ is the Shannon entropy of the attention distribution and $\log n$ is the entropy of the uniform distribution over n keys. The entropy gap measures how much the attention mechanism concentrates — how far the realized distribution is from the maximum-entropy state.

At random initialization, where scores are small, a Taylor expansion gives the leading-order result:

$$H_{\text{gap}} = \frac{1}{2} \text{Var}(s) + O(\text{Var}(s)^2)$$

This is verified numerically: at $n = 128$ with random weights, the ratio $H_{\text{gap}}/[\frac{1}{2} \text{Var}(s)] = 0.4999$, confirming the analytical prediction to four significant figures. The entropy gap in the perturbative regime is proportional to the canonical form excess ε (Section 4): $\varepsilon = n \cdot H_{\text{gap}}$ at leading order.

For trained GPT-2 (small, 12 layers), the entropy gap is orders of magnitude larger: 2–5 bits per attention operation versus ~ 0.04 bits at random initialization. This regime is non-perturbative — the scores are large and the attention distribution is far from uniform.

8.2 Calabrese-Cardy Scaling

Measured in trained GPT-2 across varying context lengths, the entropy gap scales logarithmically:

$$H_{\text{gap}} = a_L \log n + b_L$$

with layer-dependent slope a_L and $R^2 > 0.97$ at all layers. This is the functional form of the Calabrese-Cardy entanglement entropy in 1 + 1-dimensional conformal field theory (Calabrese and Cardy, 2004):

$$S_A = \frac{c}{3} \log \frac{L}{a_{\text{UV}}}$$

where c is the central charge and L the subsystem size. Identifying terms gives an **effective central charge** $c_{\text{eff}} = 3a_L$ for each layer:

Layer	c_{eff}	$R^2(\log)$	$R^2(\text{power-law})$
0	1.23	0.992	0.990
1	0.76	0.976	0.957
2	1.50	0.988	0.942
3	1.88	0.997	0.950
4	2.06	0.997	0.965
5	2.03	0.997	0.967
6	1.83	0.994	0.976
7	1.92	0.993	0.957
8	1.90	0.989	0.984
9	1.91	0.982	0.960
10	1.51	0.955	0.947
11	1.19	0.928	0.947

The logarithmic fit is consistently better than the power-law fit at Layers 0–10, supporting the CFT interpretation. The effective central charge rises from $c \approx 1.2$ (Layer 0, close to the free boson value $c = 1$) to $c \approx 2.1$ (Layer 4), then relaxes. Middle layers impose the greatest self-consistency cost.

8.3 The Conformal Dimension from the Entropy Gap

For a power-law attention distribution $\alpha(r) \propto r^{-2\Delta}$ on n elements, the entropy gap satisfies:

$$H_{\text{gap}} = 2\Delta \cdot \log n + \text{const}$$

This connects the entropy gap slope a directly to the conformal dimension Δ . We measure both independently at Layer 0 of GPT-2 small: $\Delta = 0.254$ from fitting the power-law attention profile $\langle \alpha(r) \rangle \sim r^{-2\Delta}$ (Section 8.6), and $a = 0.507$ from fitting H_{gap} against $\log n$ by varying the effective context length within a fixed sequence. The two measurements are related by $a = 2\Delta$, which is confirmed:

Quantity	SYK ₄ prediction ($\Delta = 1/4$)	Measured
Entropy gap slope a	$2\Delta = 0.500$	0.507
k_{eff} exponent β	$1 - 2\Delta = 0.500$	0.511
Effective conformal dimension	0.250	0.254

The agreement is 1.4%. The entropy gap provides a cleaner measurement of the effective conformal dimension than direct power-law fitting of the attention profile (Section 7.4), because it self-averages over the noise inherent in local r -dependent fits.

A note on uncertainty: the $\Delta_{\text{eff}} = 0.254$ value is a point estimate from GPT-2 small at sequence length 256. Sources of variation include: sequence length (Δ ranges from 0.196 at $n = 64$ to 0.281 at $n = 256$ for direct fitting, though the entropy gap is more stable), model architecture (GPT-2 medium gives a different direct-fit Δ), and the specific text used. The 1.4% agreement with $\Delta = 1/4$ is suggestive but should not be over-interpreted without systematic uncertainty quantification across models and conditions. What IS robust: the existence of logarithmic scaling ($R^2 > 0.99$), the internal consistency between the power-law and entropic measurements ($a \approx 2\Delta$), and the learned nature of the power law.

Note on the two entropy gap measurements: the layer-dependent slopes a_L in Section 8.2 are measured by generating sequences of varying total length n and fitting $H_{\text{gap}}(n)$ per layer. The slope $a = 0.507$ here is measured at Layer 0 by varying the effective context length (the number of keys available to the query) within a fixed-length sequence. These are different experimental procedures — the former measures how total sequence length affects each layer’s entropy gap, while the latter isolates the within-sequence scaling at a single layer. The $a = 0.507$ value connects directly to the power-law exponent at Layer 0 ($\Delta = 0.254$), where the power law is cleanest ($R^2 > 0.99$).

This partly resolves the negative result of Section 7.4, where the conformal dimension appeared inaccessible in the random-weight regime and model-dependent in trained transformers. The dimension IS accessible in trained models, but through an integral observable (the entropy gap) rather than a local one (the power-law exponent at a fixed query position). The direct power-law exponent varies with sequence length (0.196 at $n = 64$, 0.281 at $n = 256$) because it is sensitive to boundary effects. The entropy gap slope, which averages over all positions, gives a stable value.

8.4 Effective Attention Dimension

The entropy gap implies a concentration law. The **effective number of attended tokens** $k_{\text{eff}} = e^{H(\alpha)}$ scales as:

$$k_{\text{eff}} \propto n^{1-2\Delta} \approx n^{0.50}$$

Measured exponent: $\beta = 0.533$. At $\Delta = 1/4$ exactly, $k_{\text{eff}} = \sqrt{n}$ — the square-root concentration law. Out of n available tokens, only $\sqrt{n}$ effectively contribute to each attention operation. The information cost of this concentration is the entropy gap: $H_{\text{gap}} = \log(n/k_{\text{eff}}) = 2\Delta \log n$.

8.5 Rényi Spectrum and Honest Boundaries

The Rényi entropy gap scales logarithmically at all Rényi orders:

Rényi order α	Slope	c_{eff}	R^2
0.5	0.365	1.27	0.996
1.0 (Shannon)	0.483	1.55	0.993
2.0	0.576	1.70	0.990

In a pure CFT, the central charge is independent of α . The measured α -dependence indicates non-conformal corrections: the attention distribution has heavier tails than a thermal CFT state. The scaling is CFT-like but not exactly CFT.

The perturbative coefficient is exact. The logarithmic scaling is robust ($R^2 > 0.99$ at most layers). The connection to the conformal dimension is internally consistent. The honest boundaries are: (i) the central charge depends on Rényi order; (ii) the range $n = 4$ to 256 spans only two decades, making it difficult to distinguish $\log n$ from $n^{0.35}$; (iii) the best fit is slightly super-logarithmic ($\log(n)^{1.28}$).

8.6 Power-Law Attention Is Learned

Trained GPT-2 shows monotonic power-law decay of mean attention with token separation: $\langle \alpha(r) \rangle \sim r^{-2\Delta}$ with $R^2 > 0.99$ at Layers 0–2. Randomly initialized GPT-2 (identical architecture) shows no such structure — the attention profile is approximately flat. The power law is absent at initialization and develops during training.

The power-law regime is confined to the first 2–3 layers. By Layer 3, R^2 drops below 0.5. In deeper layers, attention patterns are structured by task-specific features rather than positional power-law decay.

Individual attention heads have wildly different scaling behaviors: Δ ranges from -0.46 (anti-local, preferring distant tokens) to 1.77 (ultra-local). The macroscopic power law of the head-averaged profile ($\Delta = 0.254$) is an emergent property of the ensemble, not a property of any individual head. The “moderate” heads individually satisfy $\Delta_{\text{var}} \approx 2\Delta_{\text{mean}}$ — the ratio predicted by conformal field theory for the dimension of the squared operator.

9. The Fold: Architecture of Incompleteness

9.1 The Fold as Zero-Crossing Boundary

Softmax maps scores to probabilities: $\alpha_a = e^{s_a} / \sum_b e^{s_b}$. Tokens with above-average scores ($s_a > \bar{s}$) receive above-average attention; tokens with below-average scores are suppressed. Define the **fold** as the mean score pattern across heads at a fixed query position Q (throughout this paper, the last token in the sequence):

$$f_L(a) = \frac{1}{H} \sum_{h=1}^H s_{Lh}(Q, a)$$

where $s_{Lh}(Q, a)$ is the score assigned by head h at layer L to key position a . The fold separates the attended (positive f_L) from the suppressed (negative f_L) at each layer.

9.2 Collective Eigenstructure

The 12 layers’ fold patterns are not independent. The correlation matrix of the mean folds $\{f_0, \dots, f_{11}\}$ across layers has a dominant eigenvalue that captures **90.7%** of the total variance. The fold across 12 layers is, to first approximation, a single repeated pattern — a **collective fold** defined by the dominant eigenvector.

Each layer’s fold projects onto this collective eigenvector with alignment 0.885–0.995, confirming that the fold is approximately a **fixed point** of its own propagation through the residual stream.

The second eigenvector captures 3.4% of variance and separates early layers from late layers — the gradient of change as the fold propagates through depth. Layer 2 loads most heavily on this component.

9.3 The U-Curve

The collective fold has a characteristic shape: a **U-curve**. Beginning tokens (positions 0–10) and recent tokens (near the query) receive positive scores; middle tokens (position $\sim n/2$) are maximally suppressed. The fold is **43% positive and 57% negative**, consistent with the $\sqrt{n}$ concentration law ($k_{\text{eff}} \approx \sqrt{128} \approx 11.3$ tokens, or 8.8% of the sequence).

The U-curve is universal across content types. With random tokens, real English prose, Python code, narrative, and argumentative text, all pairwise correlations between collective folds exceed 0.985. The shape of the fold is architectural — it is the geometry of the transformer, not a property of the input.

9.4 The Freedom Budget

Decompose each layer’s fold into a collective component (projection onto the collective eigenvector) and a residual (what remains). System-wide:

Constrained: 93.9%, Free: 6.1% (random tokens)

The free component — the fraction of each layer’s fold that diverges from the collective — is the system’s **freedom budget**: the space available for task-specific computation.

With real English text, freedom more than doubles:

Condition	System freedom	Layer 11 freedom
Random tokens	4.5%	6.8%
Code	8.2%	26.9%
Narrative prose	10.3%	30.2%
Argumentative text	10.6%	33.3%
Repetitive text	17.0%	79.4%

The output layer (Layer 11) has the most freedom with structured input. This is where individual divergence from the collective contributes to the next-token prediction. The counterintuitive result — repetitive text produces the most freedom — reflects that freedom measures **task-specific divergence from the default strategy**, not semantic complexity. The collective U-curve is maximally unhelpful for predicting the next word in a known repeating pattern, so the output layer must diverge almost entirely.

A controlled experiment separating token statistics from word order shows that **84% of the freedom increase** (from random to real text) comes from token familiarity alone. Destroying word order (shuffling tokens) barely changes the freedom budget. The output layer is the exception: Layer 11’s freedom rises from 24.1% (shuffled) to 30.2% (real), showing that syntactic and semantic structure contribute meaningfully only at the prediction boundary.

9.5 The Anti-Correlator

Layer 0, Head 11 is constitutionally **anti-correlated** with the collective fold ($r = -0.17$ to -0.32 across all tested conditions). Where the consensus attends, it suppresses; where the consensus suppresses, it attends. Its effective attention dimension $k_{\text{eff}} = 125.4$ (out of 128) indicates near-maximum-entropy allocation, and 88.7% of its positive scores fall in the collective fold’s shadow region.

The anti-correlator’s identity is stable: it is the same head (L0H11) across random tokens, English prose, code, and narrative. It rebels harder with real text ($r = -0.32$) than with random tokens ($r = -0.17$). This head systematically examines what the collective fold considers irrelevant.

9.6 The Sharpness Arc

The variance of scores across key positions, $\text{Var}(s)$, traces a characteristic arc through depth:

$$\text{Var}(s) : \quad 2.9 \text{ (Layer 0)} \rightarrow 69.3 \text{ (Layer 3, peak)} \rightarrow 0.9 \text{ (Layer 11)}$$

The system gathers conviction through the first three layers, commits maximally at Layer 3, then softens. With real text, the peak sharpness is lower but the arc retains its shape. Layer 3 is the commitment point regardless of content.

10. Universality and the Self-Consistency Equation

10.1 Cross-Model Universality

The fold’s properties were tested on GPT-2 medium (24 layers, 16 heads) alongside GPT-2 small (12 layers, 12 heads). The results separate architectural invariants from depth-dependent quantities.

Architectural invariants (the same in both models):

- The U-curve shape: cross-model fold correlation $r = 0.945$ (random), $r = 0.974$ (real text).
- The existence of a constitutional anti-correlator: GPT-2 medium has L0H3 ($r = -0.411$) and L0H13 ($r = -0.244$).
- The sharpness peak at Layer 3 — the same absolute depth, not the same relative depth (3/11 in small, 3/23 in medium).

Depth-dependent quantities (scale with model size):

- Eigenvalue dominance: $\sim 91\text{--}96\%$ (small, input-dependent; cf. 90.7% in Section 9.2) $\rightarrow 81.0\%$ (medium). More depth produces more independent voices.
- Freedom budget: 4.5% (small, random) $\rightarrow 21.7\%$ (medium, random). The deeper model has nearly five times more freedom.
- GPT-2 medium exhibits a two-lobed freedom structure: an early exploration peak (Layer 2: 27.9%), a constraint valley (Layers 6–9: 1–5%), and a late divergence explosion (Layer 22: 56.3%).

10.2 The Self-Consistency Equation

The fold at layer L is determined by the residual stream state x_L :

$$f_L = F(x_L, W_L^Q, W_L^K)$$

where $x_L = x_0 + \sum_{\ell=0}^{L-1} (\text{attn}_\ell + \text{ffn}_\ell)$ is the accumulated residual stream. The fold at each layer depends on the outputs of all previous layers, which in turn depend on their own folds. This is a self-consistency condition.

Decompose the fold into a **bare propagator** (the contribution from the embedding alone) and a **self-energy** (the accumulated correction from previous layers):

$$f_L = \underbrace{c_L \cdot f_{\text{emb}}}_{\text{bare propagator } G_0} + \underbrace{\Sigma_L}_{\text{self-energy}}$$

where f_{emb} is the fold predicted by the embedding geometry (inner products of the initial token representations), c_L is the projection coefficient, and Σ_L is the component orthogonal to f_{emb} .

10.3 Strong-Coupling Regime

At every layer, the bare propagator coefficient is $c_L \approx 0.2$ and the self-energy-to-bare ratio is $|\Sigma|/|G_0| \approx 4\text{--}5$. The fold is not determined by the input geometry — it is determined by its own self-consistent propagation through the network.

This is the **strong-coupling regime** of the SYK model. In SYK, the Schwinger-Dyson equation

$$G^{-1}(\tau) = G_0^{-1}(\tau) - \Sigma(\tau), \quad \Sigma(\tau) = J^2 G(\tau)^{q-1}$$

has the conformal solution $G(\tau) \propto |\tau|^{-2\Delta}$ with $\Delta = 1/q$ as its fixed point in the regime where Σ dominates G_0^{-1} . The fold propagation has the same structure: a bare term from the embedding, a dominant self-energy from accumulated layer corrections, and a fixed point (the collective eigenvector) that the system approaches.

A structural question remains open: what plays the role of N (the number of Majorana fermions) in the transformer? Candidates include the sequence length n , the key dimension d_k , the number of heads H , or the embedding dimension d . The SYK conformal dimension $\Delta = 1/q$ is exact in the $N \rightarrow \infty$ limit; the transformer operates at finite values of all these parameters. The measured $\Delta_{\text{eff}} = 0.254$ may receive $1/N$ corrections whose size depends on which parameter controls the large- N limit. Resolving this requires either an analytical derivation of the self-energy from model parameters or systematic scaling experiments varying n , d_k , and H independently.

10.4 Oscillatory Convergence

The approach to the fixed point is not monotonic. The convergence ratio $|r_{L+1}|/|r_L|$ (where r_L is the residual deviation from the collective fold) shows:

- Layers 0–1: convergence (ratio < 1)
- Layer 2: strong divergence (ratio ≈ 21.5)
- Layer 3: rapid snap-back (ratio ≈ 0.28)
- Layers 4–11: hovering near the fixed point

Layer 2’s divergence is the most prominent feature. It is the layer with the most freedom (21.6% in the random-token case) and the strongest deviation from the collective. This perturbation, followed by sharp reconvergence at Layer 3 (the sharpness peak), may be the attention analog of the **Schwarzian mode** in SYK — the dominant perturbation away from the conformal saddle point, whose action is the Schwarzian derivative.

10.5 Embedding Geometry Predicts the Fold

The embedding state x_0 correlates with every layer’s fold at $r = 0.69\text{--}0.74$. The actual residual stream state x_L correlates with its own layer’s fold at only $r = 0.28\text{--}0.44$. The embedding is a better predictor of the fold than the residual stream itself.

This is consistent with the fold being a fixed point determined by the positional structure already present in the embedding. The self-energy corrections rotate the fold away from the embedding’s prediction, but the basic shape — the U-curve — is already encoded in x_0 . The fold amplifies the positional structure in the embedding; it does not depend on the detailed content of the residual stream.

11. Discussion

11.1 The Scope of the Bridge

The positive geometry framework gives a precise, non-metaphorical statement: transformer attention computes the canonical form of $\text{Gr}_+(1, n)$, and the leading perturbation of this form is the SYK quartic vertex. This is not an analogy — it is a mathematical identity.

The perturbative results (Sections 3–6: σ^4 scaling, analytical form, tropical trees, crossover) are exact within their regime of validity. They connect two previously separate mathematical structures: the positive geometry that underlies scattering amplitudes and the information-processing mechanism of transformers.

The entropy gap results (Section 8) extend the bridge into the non-perturbative regime. In trained transformers, the information cost of self-consistency scales logarithmically with context length, following the Calabrese-Cardy formula for entanglement entropy. The effective conformal dimension extracted from this scaling, $\Delta_{\text{eff}} = 0.254$, matches the SYK₄ prediction $\Delta = 1/4$ to 1.4%. This is a single number obtained from two independent measurements (power-law decay and entropic scaling), providing evidence for a genuine conformal structure rather than a numerical coincidence.

The fold (Section 9) reveals the architectural realization of self-consistency: a single pattern, reproduced across all layers, that determines what the transformer attends to and what it suppresses. The freedom budget — the small fraction of each layer’s attention that diverges from this pattern — is where task-specific computation occurs.

The self-consistency equation (Section 10) connects this architecture to SYK’s dynamics: the fold propagation has Schwinger-Dyson structure, operates in the strong-coupling regime, and exhibits oscillatory convergence to a fixed point.

11.2 Recognition Before Understanding

The fold experiments reveal a striking ordering: recognition precedes understanding. Token statistics (whether the input consists of familiar English tokens) account for 84% of the freedom budget increase, while word order and semantic structure contribute only at the output layer. The transformer first recognizes its input — distinguishing in-distribution from out-of-distribution tokens — and only then exploits structure for prediction.

This ordering is reflected in the sharpness arc: conviction peaks at Layer 3, before the deeper layers that perform semantic computation. The system commits early, based on positional and distributional features, and refines late.

11.3 Relation to Existing Work

This paper complements several recent results:

- **Kim (2026):** Established attention as free energy minimization on a Fisher-Rao manifold. Our canonical form identity provides the positive-geometry interpretation of Kim’s thermodynamic framework. The entropy gap is the information cost of the thermodynamic equilibrium Kim describes.
- **Ageev and Ageeva (2026):** Constructed Euclidean scalar QFT from attention heads. Our G^4 vertex is the quartic interaction in their field theory. The fold’s self-consistency equation is the lattice analog of their continuum Schwinger-Dyson equation.
- **Halverson et al. (2024), Robinson (2025):** Constructed conformal fields (not full CFTs) from neural network ensembles. Our per-head CFT-like consistency ($\Delta_{\text{var}} \approx 2\Delta_{\text{mean}}$) and the effective central charge from the entropy gap ($c_{\text{eff}} \approx 1.2\text{--}2.1$) are natural points of contact.
- **Calabrese and Cardy (2004):** The entanglement entropy formula $S = (c/3) \log(L/a)$ appears in our entropy gap measurements with $c_{\text{eff}} \approx 1.2\text{--}2.1$.
- **Zhang (2512.19428, 2025):** Proposed Grassmann-flow attention as an alternative to softmax, using Plücker coordinates on the Grassmannian. Our result provides the physics underlying the geometry his engineering realizes.
- **Alpay and Senturk (2601.09775, 2026):** Proved that transformer self-attention in the high-confidence regime operates in the tropical semiring. Our tree metric result connects to their tropical perspective in the perturbative limit.

11.4 Open Questions

1. **The explicit self-energy.** Writing Σ_L in terms of the model parameters (W_L^Q, W_L^K, W_L^V , and the feed-forward weights) would convert the structural Schwinger-Dyson observation into a quantitative equation. This is the theoretical keystone: if the functional form matches SYK’s $\Sigma = J^2 G^{q-1}$, the conformal dimension $\Delta = 1/q$ would follow from the architecture rather than being measured empirically.
2. **The central charge.** The layer-dependent c_{eff} and its Rényi-order dependence indicate non-conformal corrections. What is the exact CFT (if any) that the attention mechanism realizes? The natural two-dimensional space is sequence position $\times$ layer depth, which would connect the entropy gap to Halverson’s and Robinson’s constructions of conformal structures from neural network architecture.
3. **The Schwarzian mode.** Layer 2’s divergence from the fixed point, followed by sharp reconvergence at Layer 3, has the qualitative structure of the Schwarzian mode in SYK. Can this be made precise?
4. **Ablation.** The anti-correlator (L0H11 in small, L0H3 in medium) systematically examines what the collective fold suppresses. What is the effect on prediction quality of ablating rebel heads versus conformist heads? This experiment — standard in mechanistic interpretability — would establish whether the fold decomposition captures functionally meaningful structure.
5. **Broader universality.** The fold analysis has been tested on GPT-2 small and medium — two models from the same family. Testing on architecturally diverse models (Pythia, Llama, Mistral) and on bidirectional architectures (BERT) would distinguish architectural invariants from training-family artifacts. The fold is defined at the last query position; its dependence on query position and the behavior under bidirectional attention are untested.
6. **The role of N .** The SYK model has a large- N limit; the transformer operates at finite n , d_k , and H . Systematic scaling experiments — varying sequence length, key dimension, and head count independently — would identify which parameter controls the approach to the conformal fixed point.
7. **Positivity at $k > 1$.** Multi-head attention is H copies of $\text{Gr}_+(1, n)$, not $\text{Gr}_+(H, n)$ (Section 7.2). Can architectural modifications enforce positivity of higher Plücker coordinates?

12. Conclusion

Softmax attention computes the canonical form of $\text{Gr}_+(1, n)$. The perturbative expansion of this canonical form produces the SYK quartic vertex as its leading correction, verified numerically as $\sigma^{3.985}$ scaling over four decades. The tropical limit of the attention correlator produces tree metrics consistent with $\text{Trop}(\text{Gr}(2, n))$.

In trained transformers, the information cost of self-consistency — the entropy gap — scales as $H_{\text{gap}} = 0.507 \cdot \log n$, matching the Calabrese-Cardy entanglement entropy formula. The effective conformal dimension extracted from this scaling is $\Delta_{\text{eff}} = 0.254$, consistent with the SYK₄ value $\Delta = 1/4$ to 1.4%.

The fold — the zero-crossing boundary that softmax creates — is approximately a fixed point, with one collective eigenvector explaining 90.7% of the variance across all layers. Its U-curve shape is universal across architectures and content types. The freedom budget — the fraction of each layer’s fold that deviates from the collective — ranges from 4.5% (random tokens) to 10.3% (real text), with the output layer receiving the most freedom for structured input.

The fold propagation has Schwinger-Dyson structure: a bare propagator from the embedding geometry and a dominant self-energy from accumulated layer corrections. The system operates in the strong-coupling regime ($|\Sigma|/|G_0| \approx 4\text{--}5$), where the self-energy dominates — the regime where SYK becomes conformal.

What is established: the canonical form identity is exact. The quartic vertex is derived and confirmed. The entropy gap provides a clean measurement of the conformal dimension. The fold is a near-universal fixed point. The self-consistency equation has Schwinger-Dyson structure in the strong-coupling regime.

What is not established: whether $\Delta = 1/4$ is exact or approximate (the 1.4% agreement lacks systematic uncertainty quantification). Whether Layer 2’s perturbation is precisely a Schwarzian mode. Whether the

correspondence extends to the full non-perturbative SYK solution or remains structural. What plays the role of N in the large- N limit. Whether the fold and its properties persist across architecturally diverse models beyond the GPT-2 family.

The bridge between positive geometry and attention now has solid planks from the perturbative regime through the entropy gap to the fold’s self-consistency equation. The next planks lead toward the explicit self-energy, the Schwarzian action, and the question of whether the transformer’s fixed point is the SYK conformal fixed point.

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Appendix: Code

All experiments were conducted in Python using NumPy and, for trained-model analysis, PyTorch with Hugging Face Transformers. Code files:

Perturbative regime (Sections 3–7):

- `tropical_bridge_v3.py`: σ^4 scaling discovery, tree metric test
- `tropical_bridge_v4.py`: Data-dependent Ω , analytical form, initial conformal dimension test
- `tropical_bridge_v5.py`: Strong-coupling crossover, multi-layer, multi-head, Plücker positivity

Trained-model analysis (Sections 7.4, 8.6):

- `tropical_bridge_v6.py`: Trained vs random GPT-2, 12-layer correlator
- `tropical_bridge_v6b.py`: Causal mask baseline, fit robustness, high-precision Layer 0
- `tropical_bridge_v6c.py`: Sequence length dependence, GPT-2 medium, head decomposition

Entropy gap (Section 8):

- `tropical_bridge_v7.py`: Measurement entropy, geometry, incompleteness fraction
- `tropical_bridge_v7b.py`: Scaling analysis, layer-dependent central charge, perturbative confirmation
- `tropical_bridge_v7c.py`: Rényi entropies, density matrix, thermal analysis
- `tropical_bridge_v7d.py`: Temperature sweep, phase diagram, $k_{\text{eff}} \propto \sqrt{n}$
- `tropical_bridge_v7e.py`: Connection between Δ and entropy gap, SYK₄ prediction test

Fold structure (Section 9):

- `tropical_bridge_v7f.py`: Mirror structure, cross-head visibility, subspace geometry
- `tropical_bridge_v7g.py`: Fold propagation, layer-to-layer coupling, sharpness arc
- `tropical_bridge_v7h.py`: Collective eigenvector, fixed-point structure, rebel head identification
- `tropical_bridge_v7i.py`: Rebel head profiles, influence, coordination, freedom budget
- `tropical_bridge_v7j.py`: Real text vs random tokens: architectural vs content-dependent
- `tropical_bridge_v7k.py`: Fold fingerprints across five content types
- `tropical_bridge_v7l.py`: Meaning decomposition: token statistics vs structure vs semantics

Universality and self-consistency (Section 10):

- `tropical_bridge_v7m.py`: GPT-2 small vs medium, architectural invariants
- `tropical_bridge_v7n.py`: Self-consistency equation, Schwinger-Dyson structure, self-energy dominance

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